# Dilan Karaguler

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## Summary

Mathematics Ph.D. candidate specializing in graph machine learning and topological data analysis, with two Intel internships building graph-based ML pipelines for semiconductor layout analysis and device classification. Experienced in turning new mathematical methods into working Python and PyTorch tools.

## Technical Skills

**Programming**: Python, R, SQL
**Methods**: Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation
**Libraries**: PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM
**Tools**: Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, LaTeX
**Domain Knowledge**: Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis

## Education

**Michigan State University** — East Lansing, MI, US
Ph.D. in Mathematics (Candidate) — August 2022 – May 2027 (Expected)
- Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

**Middle East Technical University (METU)** — Ankara, Türkiye
M.S. in Mathematics — September 2019 – September 2021
- Thesis: *A Survey on Multidimensional Persistence Theory*
- Advisor: Assoc. Prof. Semra Pamuk

**Middle East Technical University (METU)** — Ankara, Türkiye
B.S. in Mathematics, Honor Student — June 2014 – June 2019

## Industry Experience

**Software Engineer Intern** — May 2026 – August 2026
Intel Corporation — Oregon, US
- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

**Graduate Technical Intern** — May 2025 – August 2025
Intel Corporation — Oregon, US
- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

## Research Experience

**Graduate Researcher** August 2022 – Present
Michigan State University East Lansing, MI, US

- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Extended it to persistent Mayer path homology and the Mayer path Laplacian, with stability theorems and a molecular application (preprint, 2026).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

**Research Mentor** May 2023 – June 2023
Topology RTG, Michigan State University East Lansing, MI, US

- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

## Projects

**Persistent Mayer Homology with Dynamic Mode Decomposition** April 2026
Python, NumPy, SciPy, GUDHI GitHub

- Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
- Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

**Homology-Fed Neural Network Framework** November 2024
Python, PyTorch, GUDHI, NumPy GitHub

- Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
- Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

**Cellular Automata on Curved Surfaces** March 2024
Python, NumPy, Matplotlib GitHub

- Simulated Conway's Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

## Publications

**D. Karaguler** and G.-W. Wei. *Mayer Path Homology.* Foundations of Data Science, accepted (2026). doi:10.3934/fods.2026025, arXiv:2605.16525.

**D. Karaguler** and G.-W. Wei. *Persistent Mayer Path Homology and Laplacian.* Preprint (2026), to appear on arXiv.

**D. Karaguler** and G.-W. Wei. *Directed Topology for Protein–Nucleic Acid Flexibility.* In preparation (2026).

**D. Karaguler.** *A Survey on Multidimensional Persistence Theory.* M.S. Thesis, Middle East Technical University (2021).

## Leadership & Awards

**Excellence in Teaching Award**, Department of Mathematics, MSU May 2025
**Graduate Representative**, NatSci Student Advisory Council, Michigan State University 2025 – 2026
**Graduate Student Representative**, Graduate Studies Committee, Department of Mathematics, MSU 2024 – 2025
**President / Board Member**, Association for Women in Mathematics, MSU Chapter August 2022 – August 2025
**Lead Teaching Assistant**, Department of Mathematics, MSU (mentoring and observing TAs) August 2024 – Present
**Organizing Committee Member**, Graduate Student Topology and Geometry Conference April 2024